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(HONS)

ASSIGNMENT:
NETWORK SCANNING PHASE USING NMAP

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Assignment: Network Scanning Phase Using Nmap

Task 1: Host Discovery



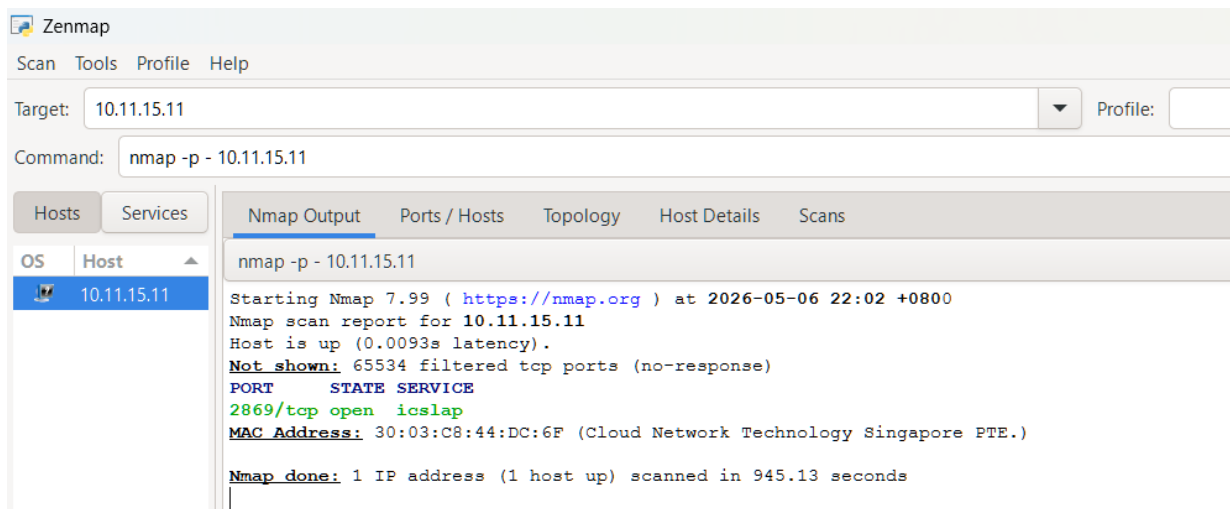
1) What hosts were discovered?

The scan successfully discovered one live host at IP address 10.11.15.11 with a very low latency of 0.0060s. The hardware was identified via its MAC address as belonging to Cloud Network Technology Singapore PTE.

2) Why is host discovery important?

It identifies which devices are active on the SayaBoleh Training Centre network so you don't waste time scanning empty IP addresses. This phase defines the scope of the security assessment and helps avoid unnecessary noise on the network.

Task 2: Full Open Port Scan



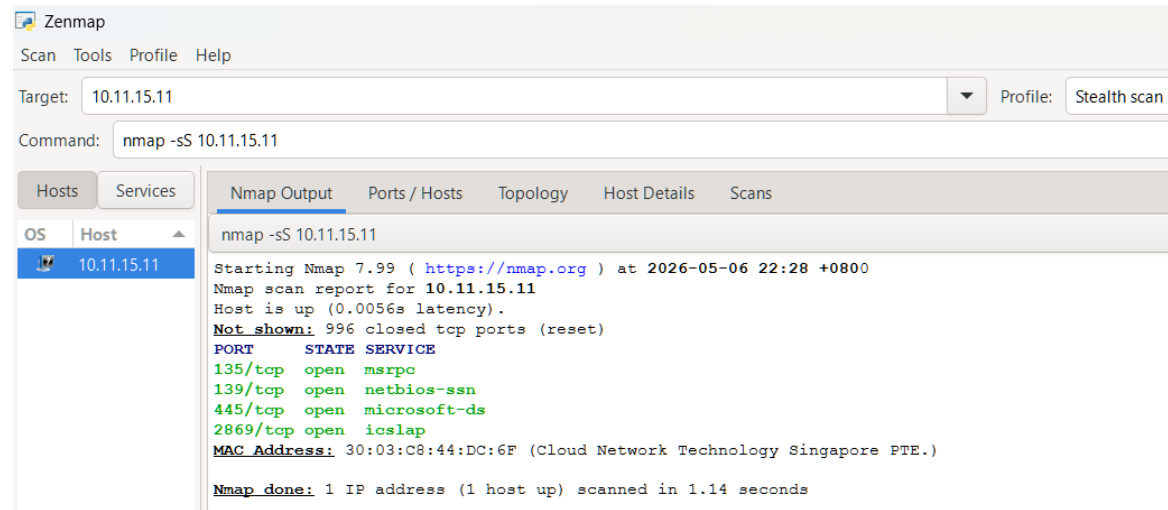
1) Open ports found

The scan identified one open TCP port: 2869. The remaining 65,534 ports were reported as filtered which means they did not provide a response due to firewall settings on the target laptop.

2) Services associated with those ports

Port 2869 is running the icslap service. In a Windows environment, this typically refers to the Internet Connection Sharing Listening Access Protocol which is often used for network discovery and UPnP (Universal Plug and Play) functionality to allow devices to find each other on the local network.

Task 3: SYN Stealth Scan



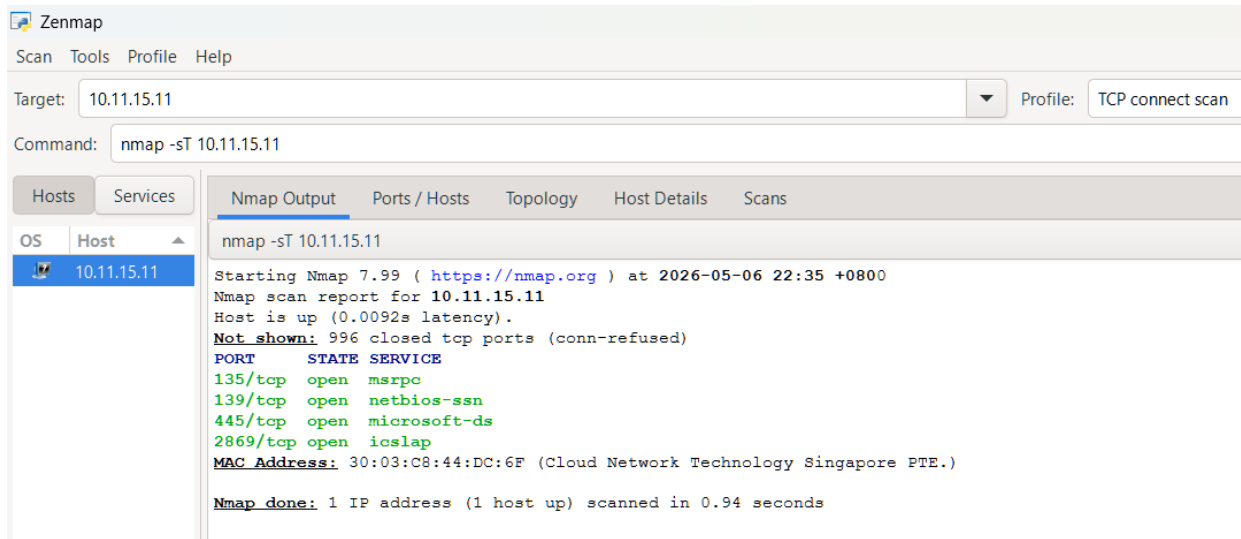
1) Why is it called stealth scan?

It is called a stealth scan because it never completes the TCP three-way handshake. It sends a SYN packet to ask if a port is open, but as soon as it receives a SYN/ACK response, it sends a RST (Reset) packet to drop the connection before it is fully established preventing many applications from logging the attempt.

2) Difference between full connect and stealth scan

A full connect scan completes the entire connection process, which is easily recorded by system and application logs. A stealth scan only performs a half-open connection, which is faster and much harder for basic security systems to detect.

Task 4: TCP Connect Scan



1) How this differs from SYN scan

Unlike the SYN scan which only performs a half-open handshake, the TCP Connect scan completes the full three-way handshake (SYN, SYN-ACK, ACK). This uses the operating system standard network functions to establish a real connection with the target. Because a full connection is made, it is much easier for the target firewalls and security software to detect and log the activity.

Task 5: Xmas Scan



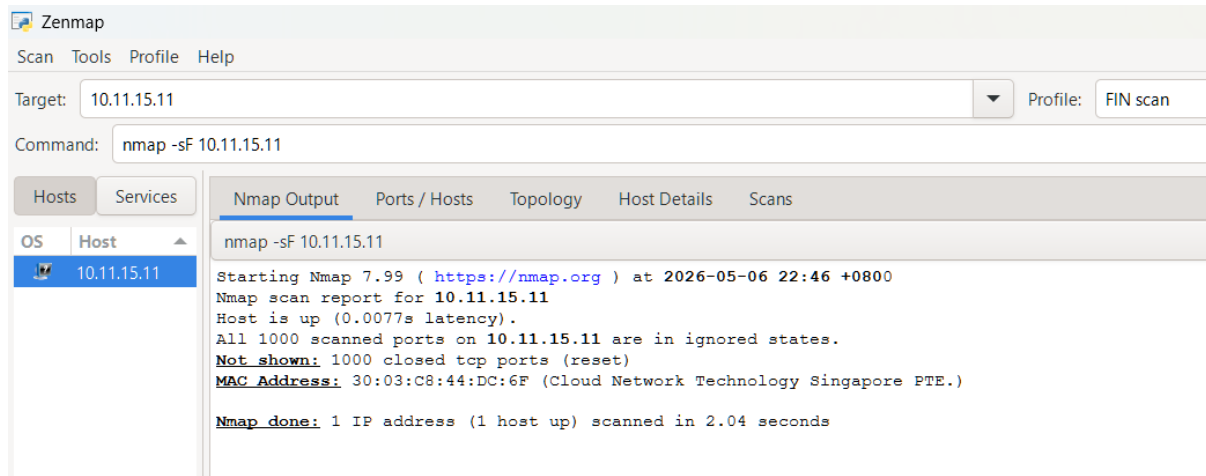
1) What packets are sent?

An Xmas scan sends a TCP packet with the FIN, PSH (Push) and URG (Urgent) flags all set to on simultaneously. This creates an unusual packet header that lights up like a Christmas tree which is where the scan gets its name.

2) Why attackers may use this scan?

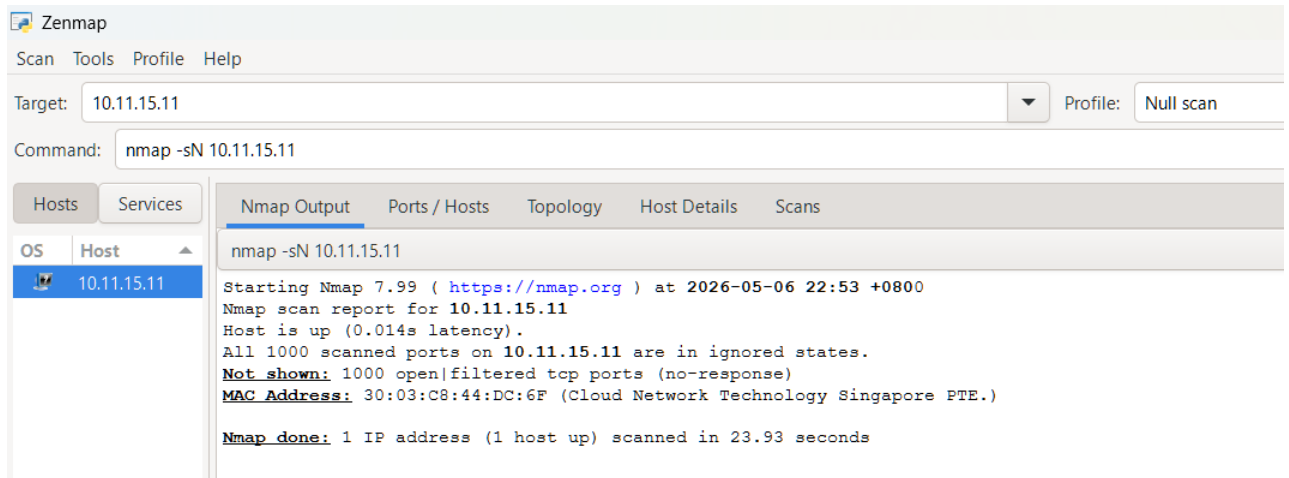
Attackers use this to sneak past non-stateful firewalls that only look for incoming SYN packets to block connections. It is also used for OS Fingerprinting as different operating systems respond differently to these illegal flag combinations. For example, Windows typically sends a RST (Reset) regardless of whether the port is open.

Task 6: FIN Scan



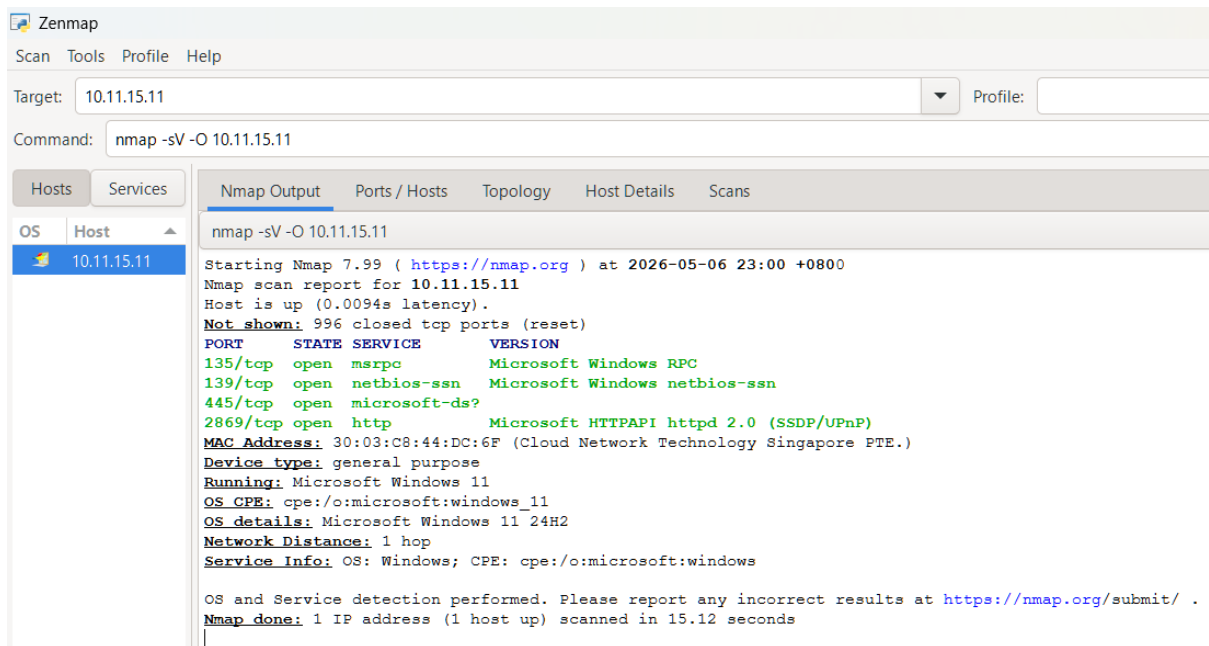
-The FIN scan on target 10.11.15.11 resulted in 1000 closed ports because the Windows operating system strictly follows the RFC 793 TCP/IP standard, which requires a RST (Reset) packet to be sent in response to any out-of-state packet like a FIN regardless of whether the port is open or closed. Because the host at 10.11.15.11 sends this uniform Reset response for every request, Nmap cannot distinguish between active services and closed ports, making the scan ineffective for port discovery. However, this specific behaviour serves as a valuable OS Fingerprinting result, as the Reset response confirms the target is likely a Windows system, whereas Linux-based systems would typically handle a FIN packet differently.

Task 7: NULL Scan



-The NULL scan performed on target 10.11.15.11 resulted in 1,000 ports being identified as open or filtered because the host Windows operating system strictly adheres to RFC 793 standards. According to this protocol, a device should send a RST (Reset) packet for any TCP segment that arrives with no flags set if the port is closed but it may ignore the packet entirely if the port is open. Because Windows often silently drops these invalid NULL packets or filters them by the native firewall, Nmap cannot definitively confirm if a port is truly open or just being blocked. While this makes the scan ineffective for precise port discovery, it provides a clear OS Fingerprinting indicator and the lack of a standard response confirms the device at 10.11.15.11 is likely a Windows machine, as other operating systems handle flagless packets differently.

Task 8: Version Detection + OS Detection



1) Service Versions

The `-sV` flag allowed Nmap to probe open ports to identify the specific software versions active on the host. The scan detected Microsoft Windows RPC on port 135, Microsoft Windows netbios-ssn on port 139, and Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP) on port 2869. Identifying these specific versions is critical for security assessments as it helps pinpoint known vulnerabilities associated with that software.

2) Possible Operating System Detected

Using the `-O` flag, Nmap performed OS fingerprinting by analyzing how the host's TCP/IP stack responded to specific probes. The scan successfully identified the target as a general-purpose device running Microsoft Windows 11, specifically version 24H2. This identification confirms the operating system and patch level of the device at 10.11.15.11.

Task 9: Reflection and Comparison

Comparison of Scan Types

- Full Connect Scan (-sT): This scan completes the full three-way TCP handshake (SYN, SYN-ACK, ACK). It is the most reliable but the noisiest because services log the completed connection.
- SYN Scan (-sS): Often called the Half-Open scan, it sends a SYN but sends a RST (Reset) immediately after receiving a SYN-ACK. It never completes the connection, making it faster and less likely to be logged by basic applications.
- Xmas Scan (-sX): This stealth scan sets the FIN, PUSH, and URG flags. It is designed to bypass stateless firewalls and identify the OS based on how the TCP stack handles illegal flag combinations.
- FIN Scan (-sF): This scan sends only the FIN flag. Like the Xmas scan, it exploits the fact that many systems respond differently to out-of-state packets, helping to bypass simple filters.
- NULL Scan (-sN): This scan sends a packet with no flags set at all. It is used to get responses from the target TCP/IP stack to identify the Operating System or bypass SYN-blocking firewalls.

Discussion

- Which is easiest to detect?

-The Full Connect Scan is the easiest to detect because it completes the full handshake, the connection is typically recorded in application logs (like web server or database logs), making it highly visible to administrators even without advanced security tools.

- Which is stealthier?

-The SYN Scan is generally considered the most practical stealthy scan for modern networks. While Xmas, FIN and NULL scans were historically stealthier against old firewalls, they are now easily flagged by modern Intrusion Detection Systems (IDS) because they use illegal or non-standard flag combinations that never occur in normal traffic.

- Which scan is most useful for defenders?

-The SYN Scan is the most useful tool for defenders and for the cybersecurity assessment because it provides a fast, accurate map of open ports without the overhead or logging interference of a full connection. However, Service/OS Detection scans (-sV -O) are the most critical for deep defense, as they reveal the specific versions and patch levels of running software, allowing defenders to identify known vulnerabilities before they are exploited.